SUPPLY CHAIN MANAGEMENT ANALYTICS DASHBOARD

BUSINESS INTELLIGENCE PROJECT | DATA ANALYTICS PORTFOLIO

Presented by Archi Dholiya

BUSINESS PROBLEM

Modern supply chains generate large volumes of operational data across suppliers, logistics, production, and sales.

However, many organizations struggle to transform this data into clear operational insights.

Without proper analytics:

- Supplier risks remain unnoticed
- Inventory inefficiencies cause capital lock-in
- Transportation delays affect customer satisfaction
- Operational costs reduce profit margins

The lack of centralized visibility leads to slow and inefficient decision-making.

PROJECT GOALS

1. Monitor supplier reliability and defect rates
2. Analyze inventory investment and stock turnover
3. Evaluate logistics performance and transportation costs
4. Understand revenue contribution across product categories
5. Build a centralized Supply Chain Intelligence Dashboard

DATASET OVERVIEW

The dataset represents a fashion and beauty supply chain operation.

- **Product Information :** Product Type, SKU, Price, Revenue Generated
- **Inventory Data :** Stock Levels, Order Quantities, Inventory Value
- **Supplier Data :** Supplier Name, Lead Time, Defect Rate
- **Logistics Data :** Shipping Times, Shipping Carriers, Transportation Modes
- **Cost Data :** Manufacturing Cost, Shipping Cost, Total Cost

TECHNOLOGIES USED

Power Bi :

- Interactive dashboards
- KPI visualization
- Multi-page analytics

SQL :

Used for data transformation and business metrics creation

Examples of engineered metrics:

- Profit
- Profit Margin
- Inventory Value
- Delivery Status , etc...

DATASET OVERVIEW

Data Engineering Tasks Performed:

- **Created business metrics :** Profit, Profit Margin, Inventory Value
- **Derived operational metrics :** Delivery Status (On-time vs Delayed)
- **Optimized performance :** Index creation for faster queries
- **Built analytical views :** Supplier Performance, Carrier Performance, Revenue Summary

PROJECT WORKFLOW

1. **Data Collection**
   Import supply chain dataset.
2. **Data Cleaning**
   Correct data types and validate metrics.
3. **Feature Engineering**
   Create operational KPIs using SQL.
4. **Dashboard Development**
   Design interactive Tableau dashboards.
5. **Insight Generation**
   Identify risks, inefficiencies, and opportunities.
6. **Business Recommendations**
   Provide data-driven operational improvements.

DASHBOARD ARCHITECTURE

1. **Executive Overview**
   High-level operational performance
2. **Supplier Intelligence Hub**
   Supplier reliability and risk analysis
3. **Inventory Intelligence Hub**
   Stock optimization and demand analysis
4. **Logistics & Delivery Hub**
   Transportation efficiency and delays
5. **Revenue & Profitability Hub**
   Financial performance monitoring

SUPPLY CHAIN ANALYTICS ARCHITECTURE

1 Data Sources :

Supply chain data collected from operational systems:

- Supplier data
- Inventory records
- Manufacturing costs
- Logistics & shipping data
- Product sales and revenue

2 Data Processing Layer :

SQL Data Engineering Key transformations performed:

- Profit calculation
- Profit margin calculation
- Inventory value calculation
- Delivery status classification
- Cost aggregation

3 Analytics Layer :

(Power BI Dashboards)

Supply chain intelligence dashboards created:

- Executive Overview
- Supplier Intelligence Hub
- Inventory Intelligence Hub
- Logistics Intelligence Hub
- Revenue & Profitability Intelligence Hub

4 Decision Layer :

The dashboard enables stakeholders to:

- Identify supplier risks
- Optimize inventory investment
- Improve logistics efficiency
- Monitor profitability

EXECUTIVE OVERVIEW DASHBOARD

EXECUTIVE OVERVIEW DASHBOARD

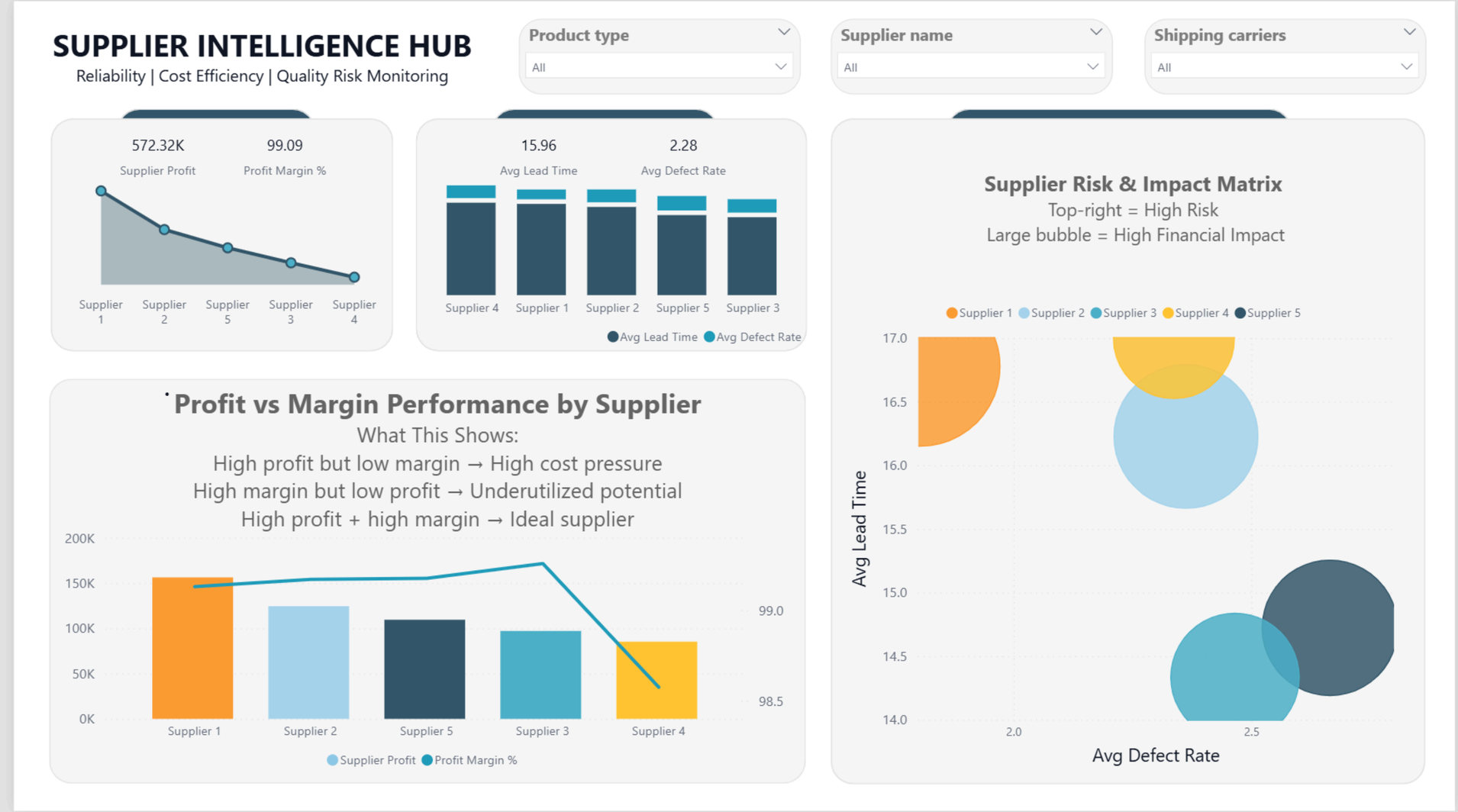
Purpose : Provide leadership with a real-time snapshot of supply chain performance.

Key Metrics : Total Revenue, Total Profit, Profit Margin, Inventory Value, On-Time Delivery Rate, Average Lead Time

Key Insight :

This page acts as the control center of the entire supply chain, enabling quick performance monitoring.

SUPPLIER INTELLIGENCE HUB



SUPPLIER INTELLIGENCE HUB

This dashboard analyzes supplier reliability and operational risk.

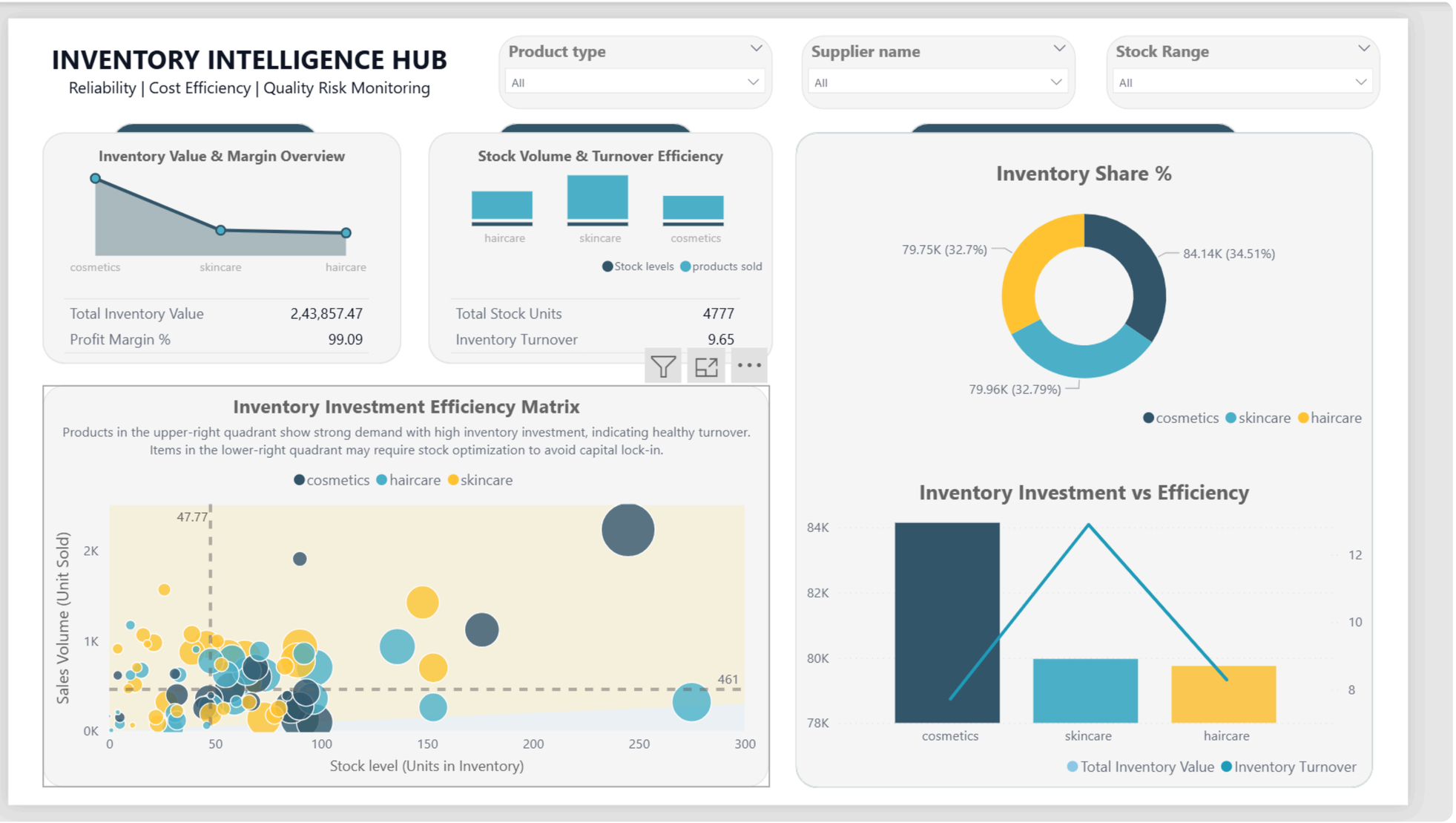
Metrics :

- Supplier Profit Contribution
- Average Lead Time
- Defect Rate
- Supplier Risk Matrix

Key Insights :

- Suppliers with high lead time and defect rate create operational risk.
- Some suppliers deliver high margins but low total profit, indicating underutilized capacity.
- Ideal suppliers demonstrate low defect rate and strong profitability.

INVENTORY INTELLIGENCE HUB



INVENTORY INTELLIGENCE HUB

This dashboard evaluates inventory investment efficiency.

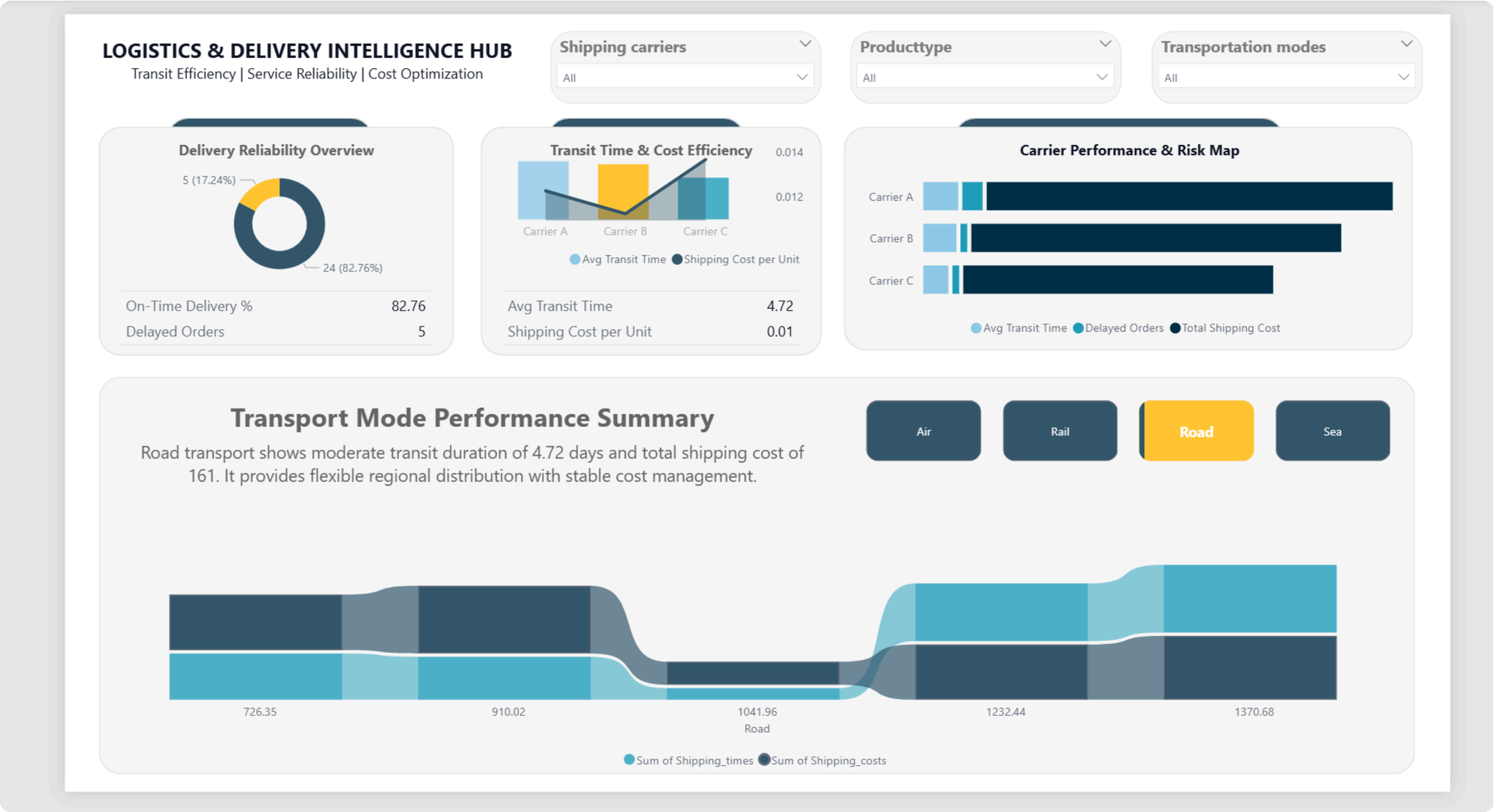
Metrics :

- Total Inventory Value
- Stock Units
- Inventory Turnover
- Inventory Share by Product Category

Key Insights :

- Products with high inventory but low sales represent capital lock-in risk.
- High sales and high stock categories indicate strong demand stability.
- Inventory turnover helps measure stock movement efficiency.

LOGISTICS & DELIVERY INTELLIGENCE HUB



LOGISTICS & DELIVERY INTELLIGENCE HUB

Focus on Transportation efficiency and service reliability

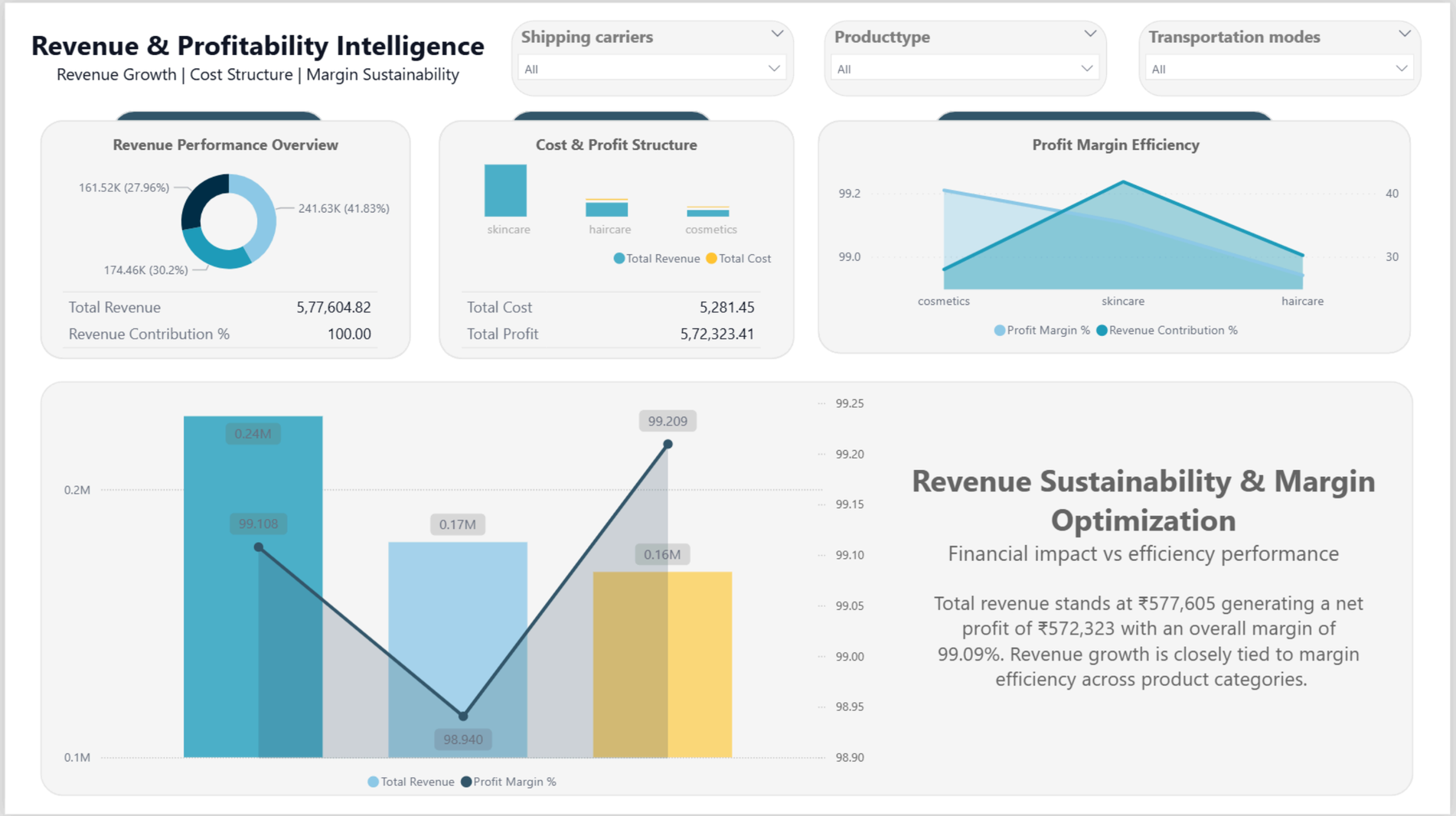
Metrics :

- Average Transit Time
- Shipping Cost per Unit
- Delayed Orders
- Carrier Performance

Key Insights :

- Some carriers show higher delay frequencies.
- Transportation mode selection directly affects delivery speed and cost.
- Logistics optimization can reduce operational delays and shipping expenses.

REVENUE & PROFITABILITY INTELLIGENCE HUB



REVENUE & PROFITABILITY INTELLIGENCE HUB

This dashboard connects supply chain performance with financial outcomes.

Metrics :

- Total Revenue
- Total Cost
- Total Profit
- Profit Margin

Key Insights :

- Skincare products generate the largest revenue share.
- Profit margins remain strong across categories.
- Monitoring cost structures helps maintain long-term profitability.

STRATEGIC SUPPLY CHAIN INSIGHTS

Insight 1 – Supplier Risk

- Suppliers with higher lead times and defect rates increase operational risk and delay product availability.

Recommendation

- Prioritize sourcing from suppliers with lower defect rates and reliable delivery performance.

STRATEGIC SUPPLY CHAIN INSIGHTS

Insight 2 – Inventory Investment

- Certain product categories maintain high inventory levels without proportional sales, creating capital lock-in.

Recommendation

- Implement demand-driven inventory planning to improve stock turnover.

STRATEGIC SUPPLY CHAIN INSIGHTS

Insight 3 – Logistics Efficiency

- Transportation modes significantly affect delivery time and cost efficiency.

Recommendation

- Optimize transportation strategy by prioritizing carriers with better on-time delivery performance.

STRATEGIC SUPPLY CHAIN INSIGHTS

Insight 4 – Revenue Drivers

- Skincare products generate the largest share of supply chain revenue.

Recommendation

- Increase inventory and supplier capacity for high-demand product categories.

PROJECT OUTCOME

This project successfully transformed raw supply chain data into actionable operational insights.

Achievements :

- Built a multi-layer supply chain analytics dashboard
- Identified supplier risk and operational inefficiencies
- Analyzed inventory investment efficiency
- Evaluated logistics performance
- Connected supply chain operations with financial outcomes

ADVANCED DASHBOARD FEATURES

This project includes several advanced analytical features.

- Interactive filters for product type, supplier, and carrier
- Drill-down analysis across dashboards
- Toggle-based insight display
  (different views depending on selection)
- Supplier risk matrix visualization
- Inventory investment efficiency matrix
- Multi-domain dashboard architecture

These features improve data exploration and decision-making capabilities.

CONCLUSION

Supply chain analytics plays a critical role in improving operational efficiency.

This dashboard demonstrates how data analytics can support smarter supply chain decisions by integrating supplier, inventory, logistics, and financial insights into a single analytical platform.

THANK YOU!!